

Ginkgolide A Prevents the Amyloid- β -Induced Depolarization of Cortical Neurons

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Supporting Information

ABSTRACT: Utilizing the *N*-methyl-D-aspartate (NMDA) receptor antagonist as a strategy, memantine is the only agent available for clinically treating mild to severe Alzheimer's disease (AD). Our aim was to develop novel similar herb-based drugs. Using a screening platform, ginkgolide A (GA), a pure compound extracted from *Ginkgo biloba*, was found to attenuate amyloid β (A β)-induced abnormal depolarization in mouse primary cortical neurons. Using receptor agonists, it was determined that GA inhibits both NMDA receptors and α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptors. Furthermore, the A β -induced increase in c-Jun N-terminal kinase phosphorylation in neurons was prevented by GA. Body weight, glutamate oxaloacetate transaminase, glutamic-pyruvic transaminase, liver histology, and kidney histology were similar when the wild-type/AD animal model mice with and without GA treatment were compared. This pure compound improves the memory of wild-type mice. Our findings indicate that GA has great potential clinically for the treatment of AD because it might target NMDA receptors just like memantine.

KEYWORDS: herbal medicines, Alzheimer's disease, ginkgolide A, amyloid β , glutamate receptors, memantine

INTRODUCTION

Ginkgo biloba, the only extant species in the division Ginkgophyta, can be viewed as a living fossil. The genome of *G. biloba* has recently been sequenced, and more than 40 000 genes have been identified in this organism, including genes that target herbivores and various pathogens.¹ Extracts of this plant have been used widely as herbal medicines to treat many diseases. However, the literature also shows that treatment with *G. biloba* results in no clinically significant benefit for patients who have diseases, such as age-related macular degeneration,² memory disorder related to multiple sclerosis,³ and various other diseases. Thus, the clinical effects of *G. biloba* in terms of the treatment of the above diseases remain controversial. Nevertheless, *G. biloba* is still widely used as a treatment in China, Germany, and France, and its most well-publicized effects involve cardiovascular and circulatory diseases. For example, EGb 761, which was extracted from *G. biloba* leaves, is able to prevent hypertension-induced oxidative stress in renal tissue.⁴ Using an animal stroke model, EGb 761 is also able to increase endogenous neurogenesis, enhance neurovascular restoration, and facilitate behavioral recovery.⁵ However, it should be noted that the use of *G. biloba* leaf extracts has also been associated with undesirable side effects, especially among individuals with blood circulation disorders and when patients are taking anticoagulants, such as aspirin or warfarin.⁶ Moreover, EGb 761 seems to have an effect on the regulation of leukocyte innate immunity because it stimulates resistance by modulating nuclear factor κ -light-chain-enhancer of activated B cells (NF- κ B) activation.^{7,8} In another study, EGb 761 was found

to reduce cell toxicity and oxidative stress by blocking mitochondria-mediated apoptosis signaling during amyloid β (A β) 25–35-induced oxidative stress and neuronal apoptosis.⁹

In addition to its major impact on the cardiovascular system, *G. biloba* also functions as an alternative medicine when treating some psychiatric disorders and the anxiety that develops with dementia.^{10,11} It has been speculated that this herbal medicine may have an effect on memory, and this has been evaluated as part of a large-scale research study called the Ginkgo Evaluation of Memory Study (GEMS).^{12–14} Recently, it seems to have been confirmed that *G. biloba* does indeed contain components that alleviate memory loss during Alzheimer's disease (AD). Ginkgolide B, a chemical extracted from *G. biloba*, has been shown to have a neuroprotective effect on A β 25–35-induced neurotoxicity in a neuroblastoma cell line.¹⁵ Similarly, EGb 761 has been found to improve cognitive function using an amyloid precursor protein/presenilin 1 (APP/PS1) animal model.^{16,17} Furthermore, impairment based on a cellular model and impairment as assessed at the animal level, which are induced by A β 1–42 or APP/PS1 gene overexpression, respectively, are able to be attenuated by YY-1224, a *G. biloba* extract that has increased amounts of a terpene trilactone present.¹⁸ These findings, when taken together, suggest that *G. biloba* might contain more

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components that could be useful when treating cognitive/memory damage.

There is an emerging understanding of the chemistry and the mechanisms of action of the compounds found in herbal plants, and this has led them to become a significant resource when exploring for new drugs. This approach also helps to compensate for the well-known problems related to the development of resistance to already available drugs. We previously established a cellular model for screening herbal plants to obtain compounds that are able to inhibit $A\beta$ -mediated glutamate receptor activation.¹⁹ We then were able to pinpoint several herbal plant crude extracts that seem to effectively alleviate $A\beta$ -induced neuronal abnormalities. In the present publication, we report that ginkgolide A (GA), a pure compound obtained from *G. biloba*, is able to attenuate $A\beta$ -induced depolarization in cortical neurons.

MATERIALS AND METHODS

Chemicals. The reagents used in this study included L-glutamine, B-27 supplement, and Neurobasal-A medium (Invitrogen/Gibco, Grand Island, NY, U.S.A.). The antibodies used in this study were against mitogen-activated protein kinase (MAPK) p42/44, MAPK p42/44 Thr202/Tyr204, p38, p38 Thr180/Tyr182, c-Jun N-terminal kinase (JNK), JNK Thr183/Tyr185, Akt, Akt Thr308, Akt Ser473, GSK3 β , and cleaved caspase-3 Asp175 (Cell Signaling Technology, Danvers, MA, U.S.A.) as well as against CaMKII and CaMKII Thr286 (Abcam, Cambridge, U.K.), GSK3 α/β Tyr279/Tyr216 (Merck Millipore, Burlington, MA, U.S.A.), anti-pro-caspase-3 antibody (Santa Cruz Biotechnology, Santa Cruz, CA, U.S.A.), anti- β -actin antibody (Sigma, St. Louis, MO, U.S.A.), and horseradish peroxidase (HRP)-conjugated secondary antibody (Thermo Scientific/Pierce, Rockford, IL, U.S.A.). Other items included GA (ChromaDex, Irvine, CA, U.S.A.), $A\beta$ 1–42 peptide (AnaSpec, Fremont, CA, U.S.A.), and bis(1,3-dibutylbarbituric acid)trimethine oxonol [DiBAC4(3)] dye (Life Technologies, Eugene, OR, U.S.A.). All other chemicals, unless indicated, were purchased from Sigma (St. Louis, MO, U.S.A.).

Animals. All animal experiments were performed with the approval of the Institutional Animal Care and Use Committee of National Taiwan Normal University (NTNU Permission 105036). C57BL/6 mice were purchased from the Academia Sinica Animal Center (Taipei, Taiwan). APP transgenic mice (J20 mice), carrying the human APP gene with the Swedish (K670N/M671L) and Indiana (V717F) familial mutations, were used as the AD mouse model.²⁰ Mice were housed with a light/dark cycle of 12 h light and 12 h dark. Food and water for mice were provided *ad libitum*. The genotyping of tail DNAs from wild-type and J20 mice was analyzed by polymerase chain reaction at around 1 month of age. Beginning at around 6 months of age, the mice were divided into four groups: wild-type mice injected with vehicle control or GA and J20 mice injected with vehicle control or GA. GA was dissolved in 50% dimethyl sulfoxide (DMSO) and was administered via subcutaneous injection at a dosage of 30 mg/kg of body weight; this was carried out 5 days a week for 4 weeks. The dosage is calculated on the basis of the half maximal inhibitory concentration (IC_{50}) and the proportion of extracellular fluid to body weight. After the period of GA and vehicle administration, the mice were subjected to various animal behavior tests. Finally, the mice were anesthetized to obtain the blood samples and then sacrificed to allow for paraformaldehyde perfusion to obtain samples from the liver, the kidneys, and various other organs. Hematology, gross pathology, and histopathology studies were also performed by the mouse core facility (Academia Sinica Mice Clinical, Taipei, Taiwan).

Mouse Primary Cortical Neurons. After mating and subsequent birth, mice postnatal days 1–3 (P1–P3) were used to obtain dissociated cerebral cortices. The protocol was adopted from Brewer and Torricelli, with a number of modifications.²¹ In brief, about 200 mg of dissociated cortex tissue was maintained by temporarily placing the tissue in cold Neurobasal-A medium. The slightly minced tissue was

then transferred to a tube and digested with 1 mg/mL papain with shaking for 20 min. After gravity precipitation, the residue in each tube had fresh Neurobasal-A medium added to it and this was followed by gentle pipetting up and down for 45 s; the tissue was then allowed settle for 2 min. This step was repeated 3 times. Each time, the supernatant was removed and collected into a single new tube. These collected cells were carefully laid onto a prepared OptiPrep gradient setup in a centrifuge tube according to the method outlined by Brewer and Torricelli.²¹ After centrifugation at 2000 rpm for 15 min, the upper layer from each tube was aspirated off and the lower layers, which contained the neuronal cells, were removed to a new tube that contained 5 mL of fresh Neurobasal-A medium. These neuronal cells were cleaned up twice more to remove gradient medium; this was performed by adding fresh Neurobasal-A medium and centrifuging at 1250 rpm for 5 min. Finally, the cells were plated onto poly-D-lysine-coated culture dishes at a density of $1.04 \times 10^5/\text{cm}^2$. The medium was subsequently changed every 3 days, with fresh Neurobasal-A medium added each time.

3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium Bromide (MTT) Assay and Cell-Based Drug Screening Assay.

Cell viability was assessed by the MTT assay. Primary cortical neurons were seeded into 24-well plates at a density of 2×10^5 cells/well. Tetrazolium dye MTT was dissolved in DMSO to make a stock solution of 5 mg/mL. After the various treatments, 50 μL of MTT stock solution was added to each well to give a final concentration of 0.5 mg/mL. After 3 h for incubation at 37 °C, the medium was replaced with 300 μL of DMSO to dissolve the formazan crystals. The plates were then shaken for 10 min at room temperature. Finally, the absorbance was measured at 570 nm using a microplate reader. Absorbance is represented as a percentage of the appropriate control, which was set as 100%.

In terms of the screening platform, this had been developed and established previously in our laboratory.¹⁹ A stock solution of dye (1 mM) was prepared by dissolving DiBAC4(3) in DMSO. Primary cortical neurons were cultivated in one well of a 4-well plate for 6 days in Neurobasal-A medium. Cells were then preincubated with herbal extracts/pure compounds for 1.5 h, followed by 0.5 h of dye incubation/buffer change. During dye incubation/buffer change, the medium of the primary neurons was replaced with the KRH buffer [127 mM NaCl, 5 mM KCl, 2 mM CaCl_2 , 1.2 mM MgCl_2 , 6 mM glucose, and 25 mM 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES)] containing a final concentration of 5 μM DiBAC4(3), herbal extracts/pure compounds, and 0.2 $\mu\text{g}/\text{mL}$ 4',6-diamidino-2-phenylindole (DAPI) or Hoechst 33342. After such a period of dye incubation/buffer change, these wells were investigated using inverted fluorescent microscopy (Carl Zeiss, Jena, Germany). Specifically, this involved the addition of 2.7 μM $A\beta$ or the appropriate treatment after which images of the cells were recorded using an excitation wavelength of 490 nm and an emission of 505–550 nm band-pass filter. The green fluorescence and blue fluorescence in the images represent the changes in membrane potential and the localization of the nuclei before treatment and at 1 min intervals after treatment up to 11 min, respectively. The intensity of the changes in green fluorescence was quantified using MetaMorph software based on the nucleus staining as the reference for cell position. We obtained a single series of images from each experiment, which was repeated 3–4 times. On the basis of the DAPI/Hoechst 33342 images and the magnification (400 $\times$), there were on average 7–10 neurons in each image. Hence, at least 20–40 neurons were quantified for each sample in total.

Y Maze. Spontaneous alternations in behavior of the mice were assessed using three equally angled arms that were 5 cm wide, 30 cm long, and 15 cm high. Each mouse was placed in one of the arm compartments and was allowed to move until its tail completely entered into another arm, which was then recorded. The sequence of arm entries was manually recorded, with the arms being labeled A, B, or C. An alternation was defined as entry into all three arms consecutively. The total number of arm entries minus 2 was used as the maximum number of spontaneous alternations that a mouse carried out. The percentage alternation was calculated as (actual alternations/maximum alternations) $\times$ 100. Each animal was allowed to explore

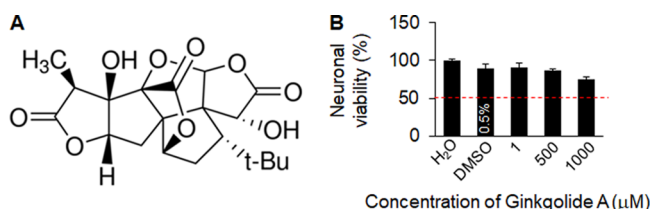


Figure 1. Structure of GA and its IC_{50} with respect to neuronal viability. (A) Structure of GA. (B) Mice primary cortical neurons were cultivated for 6 days and then treated with different concentrations of GA for 24 h. The effect of GA on neuronal viability was assessed by the MTT assay. IC_{50} of GA was estimated to be 3 mM.

the Y maze for 5 min. The apparatus was cleaned with 70% alcohol and allowed to dry between sessions.

Open Field Test. Locomotor activity in an open field was assessed using an EthoVision video tracking system and a plastic white chamber (46 × 46 × 16 cm). The surface of the chamber is rough, so that there is no reflection of the mouse being tested on the walls that might interfere with the experiment. A charge-coupled device (CCD) camera was used to detect movement in the chamber. For each animal, the open field testing was carried out for 10 min. The apparatus was cleaned with 70% alcohol and allowed to dry between sessions. An arena of 30 × 30 cm in the center was set up using the EthoVision software, and this was typically used to assess the level of anxiety.

Morris Water Maze. The water maze consisted of a pool containing water that was 120 cm in diameter. It contained a circular platform 9 cm in diameter that was submerged 1 cm below the water surface. The hidden-platform training consisted of four sessions over 4 days; each session comprised three 60 s trials with a 15 min intertrial interval. The platform location remained constant during the hidden platform sessions, but the entry points were changed semi-randomly between days. The day after the final day of hidden-platform training, testing was performed. After the testing, a probe 1 trial was conducted that involved removing the platform and allowing mice to explore the

pool for 1 min. The quadrant in which the platform was previously located was defined as the target quadrant, and the time mice spent in the target quadrant was used to measure their memory retention. At 24 h after probe 1, the time each mouse spends in the target quadrant/zone when the hidden platform was removed was tested again. This was named probe 2. The time to reach the platform and swim speed were recorded and analyzed using an EthoVision video tracking system and related software (XT 13, Noldus, Wageningen, Netherlands).

Western Blot Analysis. Cells were lysed in lysis buffer [20 mM Tris base (pH 7.9), 20 mM NaCl, 1 mM ethylenediaminetetraacetic acid (EDTA), 20 mM β-glycerophosphate, 1 mM dithiothreitol (DTT), 1 mM phenylmethylsulfonyl fluoride (PMSF), 25 nM calyculin A, 1 tablet/50 mL of protease inhibitor (Roche, Mannheim, Germany), and 0.5% Triton X-100]. The lysates were then centrifuged at 13 500 rpm for 20 min. Aliquots of each supernatant that contained equal amounts of protein, as measured by the Bradford assay (Bio-Rad, Hercules, CA, U.S.A.), were added to sodium dodecyl sulfate (SDS) sample buffer, boiled at 95 °C for 10 min, separated by sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE), transferred to a polyvinylidene fluoride (PVDF) membrane, and probed with the relevant antibody as indicated. The blots were visualized using an appropriate HRP-conjugated secondary antibody and visualized by chemiluminescence; the latter followed the protocol of the manufacturer (Millipore Corporation, Billerica, MA, U.S.A.).

Statistical Analysis. All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS) software. Differences among multiple means were assessed by one- or two-way analysis of variance (ANOVA) as appropriate, which was followed by the Dunnett's post hoc test. Some treatment effects were evaluated using two-tailed Student's *t* test and Excel software. A *p* value of <0.05 was considered to be statistically significant.

RESULTS

GA Is Able To Ameliorate Aβ-Induced Abnormal Depolarization. The structure of GA is shown in Figure 1A. We initially used vehicle alone as well as 1, 10, 100, and

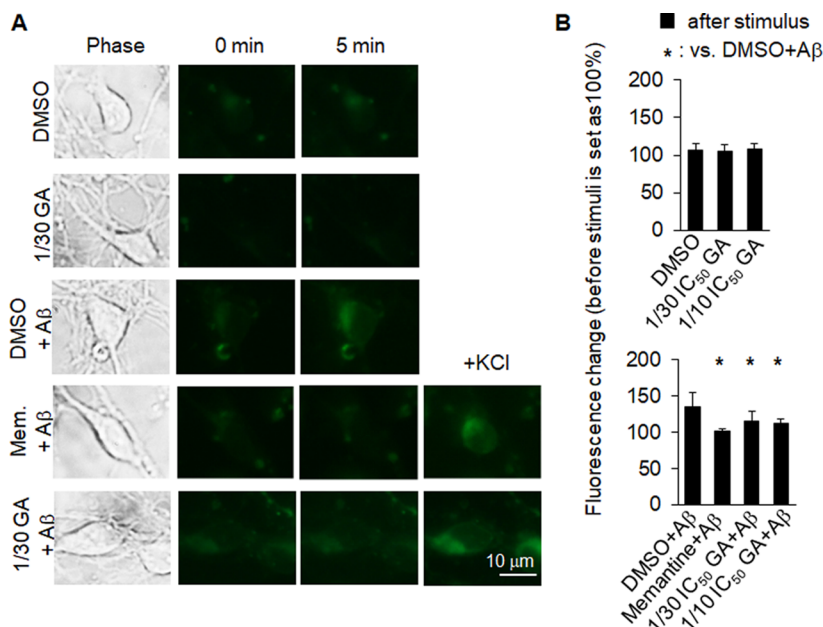


Figure 2. Aβ-induced abnormal depolarization is able to be ameliorated by the pure compound GA. (A) Primary cortical neurons that had been cultivated for 6 days were preincubated with 100 or 300 μM GA for 1.5 h in Neurobasal-A medium, followed by changing to incubate with KRH solution containing 5 μM DiBAC4(3) and 100 or 300 μM GA for an additional 30 min. Neuronal depolarization was then induced by the addition of 2.7 μM Aβ. Preincubation with 75 μM memantine was included as a positive control. Only one set of the various different doses used in the present study is shown here. (B) Intensity changes across the various groups were quantified and plotted. The images were analyzed using MetaMorph software. Data are shown as means ± standard deviation (SD) (number of neurons = 25–30). (*) *p* < 0.05 versus Aβ treatment alone.

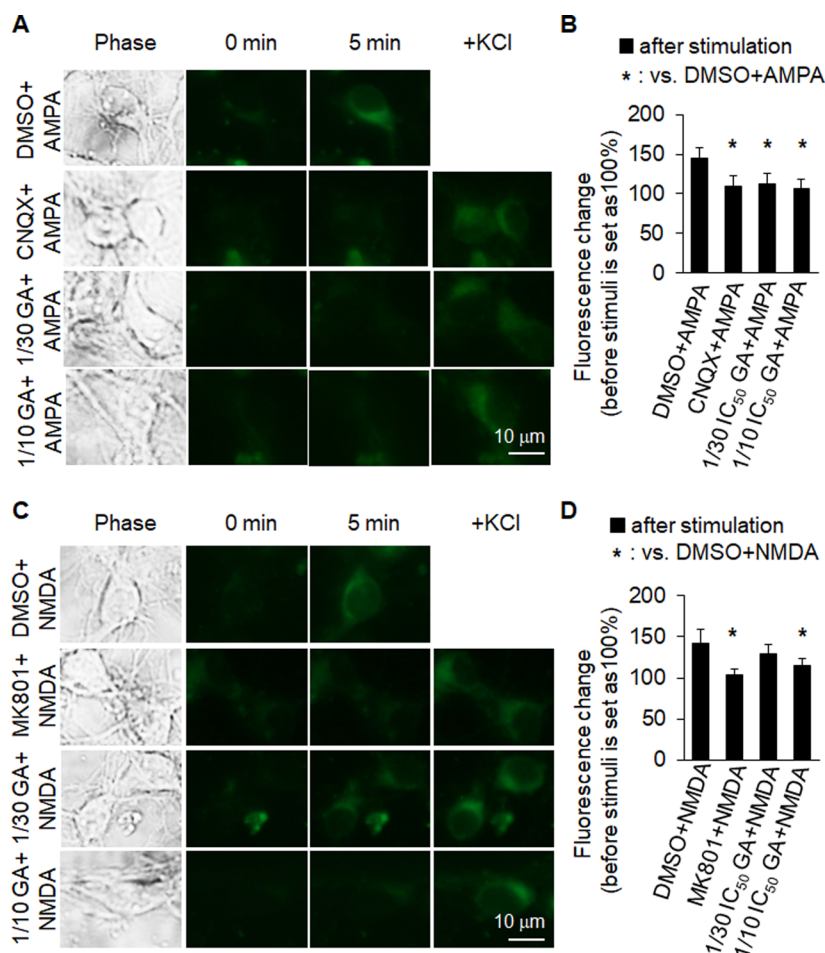


Figure 3. GA alleviates AMPA- and NMDA-induced depolarization. (A and C) Primary cortical neurons that had been cultivated for 6 days were preincubated with 100 or 300 μ M GA for 1.5 h in Neurobasal-A medium, followed by changing to incubate with KRH solution containing 5 μ M DiBAC4(3) and 100 or 300 μ M GA for an additional 30 min. Neuronal depolarization was initiated after the addition of 10 μ M AMPA or 40 μ M NMDA. KCl (45 mM) was added at the 11th min to act as a positive control to assess the responsiveness of the neurons. (B and D) Intensity changes of the different groups in panels A and C were quantified. Preincubation with 50 μ M CNQX or 100 μ M MK801 was included as positive controls. The images were analyzed using MetaMorph software. The data are presented as means $\pm$ SD (number of neurons = 25–30). (*) $p < 0.05$ versus AMPA/NMDA stimulation alone.

200 μ M GA. All of these showed no effect on neuronal survival as measured by the MTT assay. There was no significant decrease in neuronal viability until the dosage reached 500 μ M and/or 1 mM (Figure 1B). Considering the fact that GA itself might have a low level of toxicity and DMSO is used to be the vehicle, we did not use a much higher concentration to pinpoint IC₅₀. We eventually estimated IC₅₀ of GA by the MTT assay to be 3 mM on the basis of extrapolation when we are carrying out tests using primary neurons. As shown in Figure 2A, preincubation with GA at 100 μ M, one-thirtieth of the compound IC₅₀, is able to block A β -induced depolarization. Experiments were carried out using both 100 and 300 μ M GA; however, only the effect of 100 μ M is shown. The effect of memantine, a clinical drug used to treat moderate-to-severe AD patients, was examined simultaneously using the same platform; the aim was to use this as both a comparison and a control. The fluorescence intensities using GA alone, when there was incubation with A β , and when there was incubation with A β after preincubation with either memantine or GA at 100 and 300 μ M, are presented in Figure 2B. GA alone has no effect on the membrane potential of the cortical neurons. Treatment with A β significantly increases the fluorescence

intensity by about 1.5-fold. This change was inhibited by preincubation with either memantine or GA at 100 or 300 μ M.

GA Ameliorates α -Amino-3-hydroxy-5-methyl-4-isoxazolepropionic Acid (AMPA) and N-Methyl-D-aspartate (NMDA)-Induced Neuronal Depolarization. We used agonists targeting AMPA receptors (AMPA) and NMDA receptors (NMDARs) to further dissect the mechanism by which GA acts. When the neurons were pretreated with various concentrations of GA for 2 h followed by AMPA treatment, AMPA-induced depolarization was prevented (Figure 3A). Quantification shows that GA at both 100 and 300 μ M has this effect (Figure 3B). CNQX, an antagonist of AMPAR, was able to block the AMPA-induced depolarization when this was used as a control. Similarly, the images in Figure 3C show that preincubation with GA is able to inhibit NMDA-induced depolarization. The quantification results show that GA at 300 μ M but not at 100 μ M has this effect (Figure 3D). Finally, MK801, an antagonist of NMDAR, was used as a control to demonstrate the inhibition of NMDA-induced depolarization using this platform.

GA Ameliorates A β -Induced Neuronal JNK Phosphorylation. To investigate the molecular mechanism of GA, a

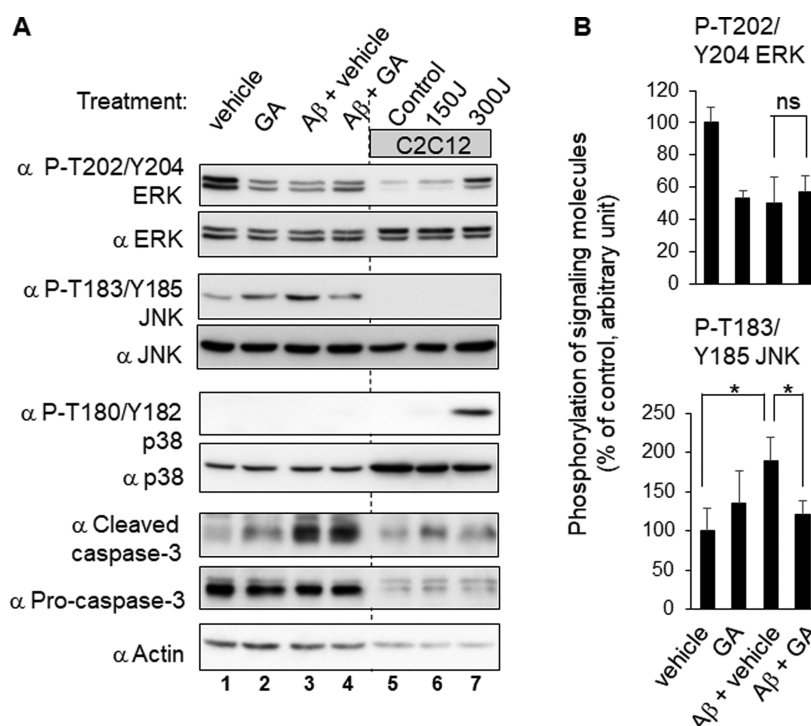


Figure 4. GA ameliorates A β -induced neuronal JNK phosphorylation. (A) Primary cortical neurons were cultivated for 6 days and then preincubated with vehicle or 10 μ M GA for 2 h, which was followed by 2.7 μ M A β treatment for 9 h. The cell lysates then underwent western blot analysis to detect various signaling molecules. Lysates from C2C12 myoblasts treated with UV irradiation were used as antibody controls. These blots are representative of an experiment that was repeated 3 times. (B) Phosphorylation of Erk and JNK was quantified from the blots using National Institutes of Health (NIH) ImageJ freeware. The y axis represents the percentage of the control values (lane 1), and this was set to 100%. Data are shown as the means $\pm$ SD ($n = 3$). (*) $p < 0.05$. ns = not significant.

number of possible signaling pathways that have been documented in the literature were examined. When primary neurons were preincubated with 10 μ M GA for 2 h, A β -induced JNK phosphorylation was reversed but no changes in a range of other signaling molecule levels were found (Figure 4A). The ability to detect changes in these signaling molecules was confirmed using ultraviolet (UV) radiation of C2C12 cells, a mouse skeletal muscle cell line, with lanes 5–7 acting as positive controls for the antibodies. Quantification of the above blots from three independent experiments was carried out, and the results are presented in Figure 4B. The findings indicate that A β induces an increase in JNK phosphorylation and that this is significantly inhibited by GA; however, in the same circumstances, GA has little or no effect on Erk/p38 phosphorylation, caspase-3 cleavage, or any other of the tested signaling pathways, which includes those involved in the Akt, GSK3, and CaMKII pathways (Supplemental Figure S1 of the Supporting Information).

GA Does Not Affect Various Physical and Physiological Parameters. First, the presence of basic level toxicity that would affect the physiological status of the animals and their tissues was examined before we could begin to interpret the efficacy of GA using an animal model. The flowchart of the experimental design was depicted in Figure 5A. As shown in panels B, C, and D of Figure 5, the body weight, glutamate oxaloacetate transaminase (GOT), and glutamic-pyruvic transaminase (GPT) results showed no difference between the group with and without GA injection, respectively. We also examined the tissue morphology of liver samples (Figure 5E) and kidney samples (Figure 5F), and the histological findings showed no identifiable abnormalities affecting these tissues.

GA Has No Effect on the Y Maze and Open Field Test Results Using Either Wild-Type or J20 Mice but Does Produce an Improvement in the Memory Behavior of the Wild-Type Mice but Not J20 Mice.

First, we examined the spontaneous memory and anxiety of the two groups of mice via the Y maze and the open field test, respectively. Although the J20 mice did show a defect that affected both behavior tests compared to wild-type mice, GA given to the J20 mice resulted in no improvement in Y maze results (Supplemental Figure S2 of the Supporting Information) nor the open field test results (Figure 6A). A representative set of traces is shown in Figure 6A. The quantification of these results shows that the time spent in the center area of the open field do not seem to be different after GA treatment when wild-type or J20 mice are used (Figure 6B). As shown in Figure 6C, the distance moved as an index of locomotor activity showed the same pattern for the wild-type and J20 mice. We further examined the effect of GA on the AD animal model J20 mice using the Morris water maze. Before training, the latency to reach the hidden platform showed a statistically significant difference between the wild-type and J20 mice at 7 months old (session 1 in Figure 6D). However, there was no difference in this parameter between the wild-type mice and the J20 mice after four sessions of training. Testing on day 5 shows that GA also has no effect on the wild-type or J20 mice (Figure 6E). Probe 1 is believed to test the short-term memory of the mice immediately after the testing, and this is performed by removing the hidden platform. Interesting, the time spent in the target quadrant during probe 1 for the wild-type mouse group that had been injected with GA was found to be significantly greater; thus, the treated wild-type mice spent more

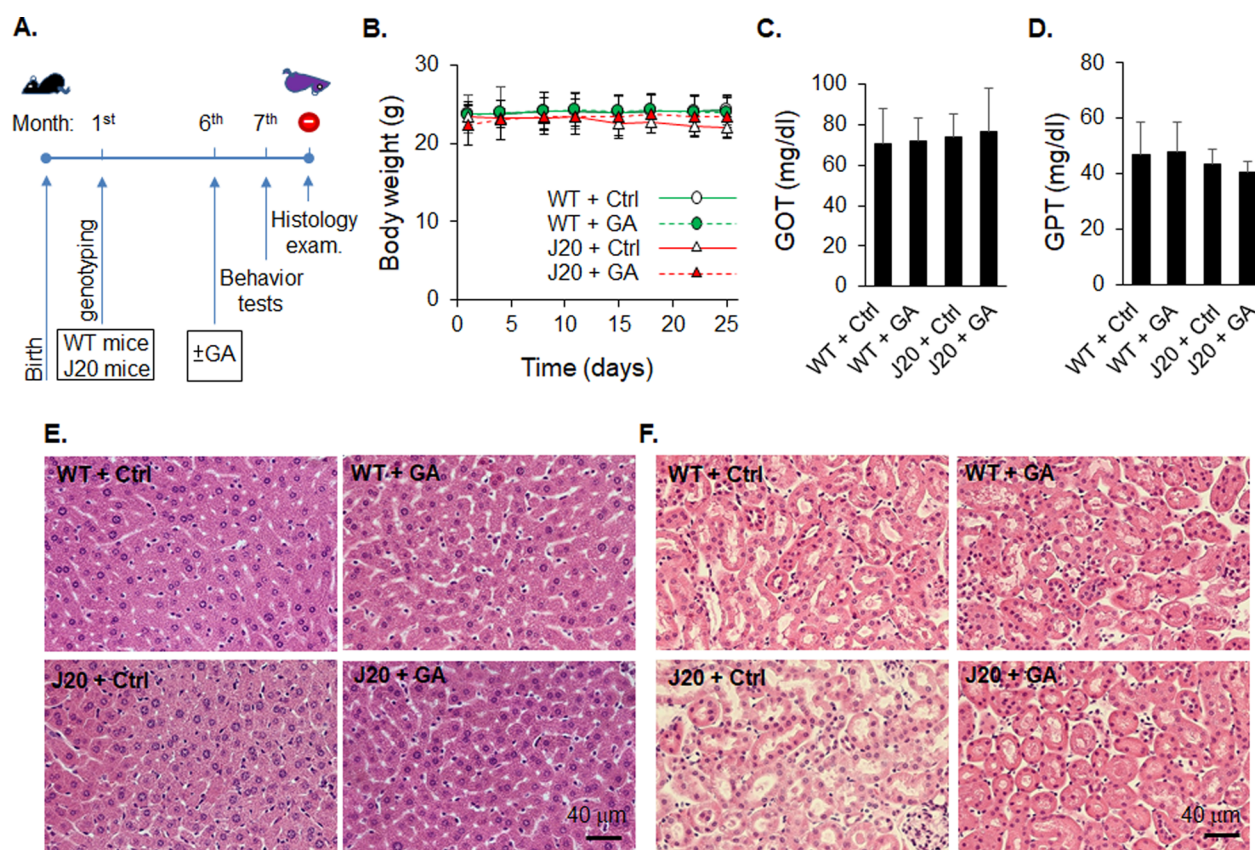


Figure 5. Assessment of the GA toxicity on both wild-type and J20 mice. (A) Flowchart of experimental procedures. The genotyping of tail DNAs from wild-type and J20 mice was analyzed by polymerase chain reaction at around 1 month of age. These mice were injected with vehicle or GA at around 6 months of age. A dosage of 30 mg/kg of body weight 5 days a week for 4 weeks was administrated via subcutaneous injection. At around the 7th month, mice were subjected to the animal behavior tests. These mice were then anesthetized and sacrificed to obtain samples from blood, liver, kidneys, and other organs. The hematology, gross pathology, and histopathology studies were analyzed. (B) Body weights during the period of GA administration, (C) concentrations of blood GOT, and (D) concentrations of GPT of the mice were measured. After sacrifice, tissue samples from (E) liver and (F) kidneys were subjected to hematoxylin and eosin (H&E) staining.

time searching for the platform in the target quadrant compared to the vehicle injection group (Figure 6F). However, no such effect was seen with the J20 mice. At 24 h after this test was performed, the same experiment was performed at second time, and the results are shown in Figure 6G; this is a test that assesses the long-term memory of the mice. The results show that the improvement effect caused by GA on wild-type mice still seems to be present. The speed of movement of the tested mice shows no difference in terms of its mean (Figure 6H).

DISCUSSION

On the basis of information from a database (<http://tcm.cmu.edu.tw/zh-tw/about01.php?menuid=1>), we tested a number of pure compounds known to be present in *G. biloba*, with this species having been screened as a positive result by our drug screen platform. GA, a pure compound found in *G. biloba*, has not been documented in the literature as having an effect on the cellular mechanisms known to be involved in A β -induced neuronal depolarization nor has it been tested using any AD mice model. We found that GA is capable of ameliorating A β -induced abnormal depolarization. GA seems to affect both NMDAR and AMPAR as well as perhaps other related pathway(s) to bring about this inhibitory effect. Further investigation revealed that this compound was able to alleviate the A β -induced pathophysiological status in neurons via a decrease in JNK phosphorylation. Although the timing and dosage given

were found to show no significant effect when tested using the AD animal model J20 mice, GA did clearly bring about an improvement in the memory behavior of wild-type mice.

As mentioned above, whether or not *G. biloba* is able to prevent dementia still remains moot. Nevertheless, very little is known in the literature regarding the effects of GA, a chemical present in *G. biloba*, at the cellular level and/or using an AD animal model. The majority of previous relevant studies have focused on aspects of disease and biochemistry other than AD. For example, GA is able to inhibit nitric-oxide-induced neurotoxicity,²² attenuate homocysteine-induced endothelial dysfunction in porcine coronary arteries,²³ and inhibit vascular smooth muscle proliferation and migration *in vitro*.²⁴ In particular, GA is known to greatly reduce the lipopolysaccharide-induced inflammatory response.^{25,26} In terms of metabolism, GA is able to ameliorate the pathological changes brought about by stimulation with high glucose²⁷ or a high-fat diet.²⁸ One report does investigate the effect of GA on AD, and this was published by Chen et al.²⁹ They examine the effect of GA on okadaic-acid-induced tau hyperphosphorylation and found that GA is able to increase cell viability and suppress the phosphorylation of tau and GSK3 β in cell lysates. On the other hand, in their system, the phosphorylation of PI3K and Akt were promoted by GA. Overall, the above reports reveal that GA has multiple effects, many of which might be beneficial. Nevertheless, our research is the first to report that GA might

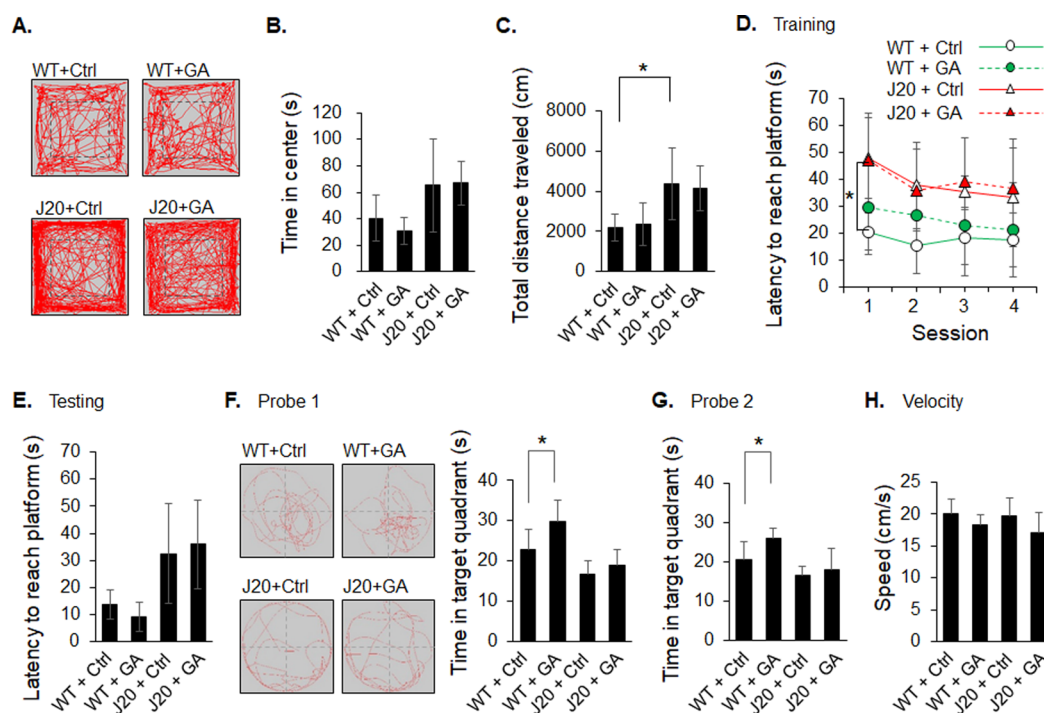


Figure 6. GA shows no effect on the open field tests but improves the memory behavior of wild-type mice while not affecting J20 mice. (A) These mice were subjected to the open field test. Mice were placed in the 46 × 46 × 16 cm plastic white chamber with a CCD camera used to detect movement in the chamber. For each animal the open field testing was carried out for 10 min. The insets are representative traces. (B) Arena 30 × 30 cm was set up using the EthoVision software to calculate the time spent in the center. The y axis represents the time spent in the center. (C) y axis represents the distance traveled. Data are shown as the means ± SD ($n = 6-7$). (*) $p < 0.05$. (D) For the Morris water maze test, mice were trained one session per day for 4 days. The parameter involved measuring the time that each mouse needed to reach the hidden platform under water during the training session. (E) One day after the final day of hidden-platform training, testing was performed. The mice were tested using the time to reach the hidden platform. (F) To examine the short-term memory of the mice, the experimental animals were tested on the basis of the time they spend in the target quadrant/zone when the hidden platform was removed right after the testing test. This was named probe 1. The insets are representative traces. (G) At 24 h after probe 1, the time that each mouse spends in the target quadrant/zone when the hidden platform was removed was tested again. This was named probe 2. (H) Distances divided by the time were also plotted. All data are shown as the means ± SD ($n = 5-7$). (*) $p < 0.05$.

potentially be a treatment for AD based on our cellular model.

GA is able to inhibit $A\beta$ -induced and NMDA/AMPA-induced depolarization, which implies that there are two possibilities in terms of a molecular mechanism. One is that GA could directly target the $A\beta$ molecules, thus preventing the activities of $A\beta$. However, Guo et al. has shown that GA is a one of many compounds/extracts that do not bind to $A\beta$ using an *in vitro* binding assay.³⁰ This result makes this possible mechanism much less likely. The other possibility is that GA might act as an inhibitor of either or both NMDAR and AMPAR via an antagonistic effect. Memantine is an uncompetitive NMDAR antagonist and is able to inhibit NMDAR channels when such receptors are in an excessively excitotoxic condition.³¹ In this context, the functional similarity between GA and memantine might indicate there is also a structure relationship, for example, in terms of functional groups. Nevertheless, how GA antagonizes these receptors needs further investigation.

In addition, the current study reveals that the inhibitory effect of GA on $A\beta$ -induced pathophysiology involves a reversal of JNK phosphorylation. We have previously reported that $A\beta$ induces an increased phosphorylation of JNK and that this seems to be reversed by treatment with either memantine or baicalein.¹⁹ These results indicate that JNK or a related signaling pathway might also be worth targeting in future studies. This might provide multiple/alternative strategies for

identifying molecules for drug development in the context of AD treatment.

It is worth taking note that the concentrations of GA (100 and 300 μM) used in depolarization experiments might seem high. Nevertheless, this system also depends upon an optimal concentration of $A\beta$. To observe a significant change in fluorescence using $A\beta$ in our screening platform *in vitro*, 2.7 μM $A\beta$ was added, which induces a strong signal. Even memantine, the clinical drug used to treat AD, is used in the range of 50–100 μM to see its effect when ameliorating $A\beta$ -induced depolarization in our platform. Actually, we did use a low dosage, 10 μM , when studying the effect of GA on the signaling molecules but not during the depolarization experiments. This was because of the different treatment durations in these experiments. When we adopted the low dosage of GA, we found that such a dose of GA is still able to ameliorate $A\beta$ -induced JNK phosphorylation.

GA is able to alleviate the depolarization induced by $A\beta$ in a neuronal cell system but is unable to improve memory behavior when tested using an AD animal model, namely, J20 mice. In this context, there are several aspects of this observation that need to be addressed. First, a study of the pharmacokinetics and bioavailability of GA and ginkgolide B using beagle dogs has revealed that the free form of these prototype ginkgolides and their absolute bioavailabilities are high.³² This suggests that administration of ginkgolides is able to promote the *in vivo* exposure to and *in vivo* bioavailability of GA. Second, there is

evidence to support the idea that GA might find it difficult to pass through the blood–brain barrier.^{33,34} Thus, this becomes one possible reason why GA is not able to boost the memory behavior of J20 mice. In this context, it is quite possible that there are differences in the blood–brain barrier between the wild-type and J20 mice. Finally, it is worth noting that J20 mice begin to show defective behavior as early as 6 months old. This was the rationale by which we chose the timing used to give GA to the mice in this study. However, the dosages and timing chosen by us might still be inappropriate for the J20 mice. Because the effects at an animal level are not as yet conclusive, the question remains as to how to overcome the blood–brain barrier using other dosages/timings/durations. Alternatively, perhaps another animal model might be found to be more appropriate when testing in the future. On the other hand, the beneficial effect of GA on the wild-type memory might also be indirect. One possibility is that GA boosts circulation in the brain, and this has an effect on memory. Whatever is the mechanism by which GA acts, it does indeed have an effect that enhances the memory of the wild-type mice. Judging from our findings, GA needs to be further investigated and developed as a potentially beneficial health product.

When our findings are taken as a whole, we have found that GA is able to ameliorate A β -induced abnormal depolarization. As shown in Figure 7, the inhibition by GA seems to involve

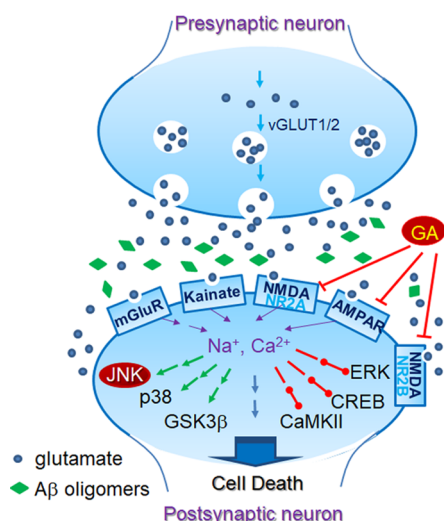


Figure 7. Plausible mechanism for the inhibition of A β -induced depolarization by GA. It has been shown that A β increases the glutamate concentration within the cleft of synapses to cause abnormal depolarization. The alleviation of A β -induced depolarization by GA might be mediated via NMDAR and AMPAR. This could further affect a JNK-related signaling pathway but not the other signaling pathways that we examined. The outcome could then affect the neuronal cells responding to the stress/death that results from the presence of A β .

both the NMDA- and AMPA-induced pathways. This implies that it acts on both NMDAR and AMPAR. Furthermore, GA alleviation of the A β -induced pathophysiological status of the cells is related to a decrease in JNK phosphorylation. Although the timing and dosage given to the AD animal model shows no beneficial effect, it is still able to bring about an improvement in the memory behavior of wild-type mice. On the basis of the current findings at a cellular level, GA remains a strong candidate for the development of potentially useful clinical drugs for AD treatment in the future.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jafc.8b04514.

GA ameliorates A β -induced neuronal JNK phosphorylation but not Akt/GSK/CaMKII signaling (Supplemental Figure S1) and GA shows no effect when the mice are tested using a Y maze (Supplemental Figure S2) (PDF)

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Author Contributions

Li-Chen Kuo and Yan-Qing Song designed and performed the experiments as well as assembled, analyzed, and interpreted the data. Chien-An Yao and Irene H. Cheng provided essential materials and interpreted the data. Chiang-Ting Chien, Guan-Chiun Lee, and Wen-Chin Yang analyzed and interpreted the data. Yenshou Lin designed the experiments, interpreted the data, supervised the study, and wrote the manuscript. All of the authors have read and approved the manuscript.

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Notes

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